

Purpose

To verify that software meets business requirements and to eliminate defects before release. IBM (2025) states, "The testing phase begins after the development team has created a functional piece of software. During this phase, the team looks for opportunities to eliminate bugs and enhance the final product."

Types of Testing

- **Unit Testing:** Validates individual components in isolation (testRigor, 2026).
- **Integration Testing:** Tests how modules interact when combined (Sahi Pro, 2025).
- **System Testing:** Evaluates the entire, integrated application against defined requirements.

Testing Tools

- **Selenium:** Automates browser interactions for functional web testing.
- **JMeter:** Measures system performance and behavior under specific load levels.
- **Postman:** Automates API testing to ensure correct data flow.

Automated Testing Efficiency

Automation removes repetitive manual tasks, reducing human error and speeding up feedback. KMS Technology (2024) notes, "By automating repetitive tasks, teams can streamline workflows, reduce the margin for error, and significantly enhance overall software quality."

Case Study

ASOS utilized load testing to simulate high-traffic conditions, allowing developers to identify and optimize infrastructure bottlenecks *before* they caused real-world site crashes (The Knowledge Academy, 2026).

References

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